



December 2025

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Playfair cipher is an example of _____**
A) mono-alphabetic cipher
B) substitution cipher
C) transposition cipher
D) additive cipher
2. **The DES algorithm has a key length of**
A) 128 Bits
B) 32 Bits
C) 64 Bits
D) 16 Bits
3. **What is the primary goal of a Man-in-the-Middle (MITM) attack?**
A) Unauthorized access to sensitive information
B) Encryption of files for ransom
C) Intercepting and manipulating communication between two parties
D) Disrupting network services
4. **Which systems use a timestamp?**
i) Public-Key Certificates
ii) Public announcements
iii) Publicly available directories
iv) Public-Key authority
A) i) and ii)
B) iii) and iv)
C) i) and iv)
D) iv) only
5. **Which of the following is not an element/field of the X.509 certificates?**
A) Issuer Name
B) Serial Modifier
C) Issuer unique Identifier
D) Signature
6. **In a Caesar Cipher, if the key is 4, then the plaintext "DATA" becomes:**
A) HEHE
B) HEVX
C) HETE
D) JDXA
7. **How can a Distributed Denial of Service (DDoS) attack impact a web application?**
A) By improving website aesthetics
B) By increasing server performance
C) By overwhelming server resources and making the application unavailable
D) By designing efficient database structures
8. **In the RSA algorithm, we select 2 random large values 'p' and 'q'. Which of the following is the property of 'p' and 'q'?**
A) p and q should be divisible by $\Phi(n)$
B) p and q should be co-prime
C) p and q should be prime
D) p/q should give no remainder

CS3222-1

9. In tunnel mode, IPsec protects the _____.
 A) Entire IP packet
 B) IP header
 C) IP payload
 D) IP trailer
10. Cryptanalysis is used _____.
 A) to make new ciphers
 B) to encrypt the data
 C) to find some insecurity in a cryptographic scheme
 D) to increase the speed
11. For $p = 11$ and $q = 19$ and choose $e = 17$. Apply RSA algorithm where message = 5 and find the cipher text.
 A) $C = 92$
 B) $C = 80$
 C) $C = 56$
 D) $C = 23$
12. What security benefit does SSL/TLS encryption provide for online transactions?
 A) Improving website aesthetics
 B) Encrypting user credentials and financial information during transactions
 C) Enhancing server performance
 D) Granting unrestricted access to all users
13. For each _____ the Kerberos Key Distribution Center (KDC) maintains a database of the realm's principal and the principal's associated "secret keys".
 A) key
 B) realm
 C) document
 D) none of the mentioned
14. In the DES algorithm the round key is _____ bit and the Round Input is _____ bits.
 A) 48, 32
 B) 64, 22
 C) 66, 24
 D) 38, 52
15. Which systems use a timestamp?
 i) Public-Key Certificates
 ii) Public announcements
 iii) Publicly available directories
 iv) Public-Key authority
 A) i) and ii)
 B) iii) and iv)
 C) i) and iv)
 D) iv) only
16. The subject unique identifier of the X.509 certificates was added in which version?
 A) 1
 B) 2
 C) 3
 D) 4
17. In tunnel mode, IPsec protects the _____.
 A) Entire IP packet
 B) IP header
 C) IP payload
 D) IP trailer
18. For $p = 11$ and $q = 19$ and choose $d = 17$. Apply RSA algorithm where Cipher message = 80 and thus find the plain text.
 A) 54
 B) 43
 C) 5
 D) 24
19. Which component is included in IP security?
 A) Authentication Header (AH)
 B) Encapsulating Security Payload (ESP)
 C) Internet key Exchange (IKE)
 D) All of the mentioned
20. Diffie-Hellman key exchange is vulnerable to:
 A) Ciphertext-only attack
 B) Chosen ciphertext attack
 C) Man-in-the-middle attack
 D) Brute force attack

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the important design principles of block ciphers.	6	L*2	1	1
b)	Compare public-key cryptography and symmetric-key cryptography in terms of key management, complexity, speed and security. Evaluate which method is more effective for secure communications over the internet and justify your answer.	6	L2	2	1

c)	Differentiate between Monoalphabetic and Polyalphabetic ciphers with examples.	4	L2	1	1
2. a)	Given two prime numbers $p=11$, $q=19$ and considering $e=17$, perform the RSA encryption and decryption for the message $M=5$.	6	L3	2	1
b)	Explain the following:				
i.	One time pad				
ii.	Avalanche effect	6	L2	2	1
c)	Encrypt the plaintext "SUCCESS" using the Playfair cipher with keyword "KEYWORD".	4	L3	1	1
3. a)	Explain the working of DES encryption algorithm with neat diagram.	6	L2	1	1
b)	Encrypt the plaintext COMPUTER and the keyword HILL using Hill cipher and show the resulting cipher text.	6	L3	1	1
c)	Explain the symmetric cipher model with a neat diagram.	4	L2	1	1

Unit – II

4. a)	Evaluate a Diffie - Hellman key exchange concept for prime number $q = 71$ and primitive root $a = 7$.				
i)	If user A has private key $X_A = 5$, what is A's public key Y_A .				
ii)	If user B has private key $X_B = 12$, what is B's public key Y_B .				
iii)	What is the shared key?				
	Also explain briefly how a man-in-the-middle attack could compromise this key exchange.	8	L3	3	1
b)	Explain the Kerberos authentication protocol and illustrate the message flow between the client, Authentication Server (AS), Ticket Granting Server (TGS), and service provider.	8	L2	4	1
5. a)	Explain the Elliptic Curve encryption and decryption process for the analogy of Diffie Hellman Key exchange with neat steps. Also explain the security of ECC.	8	L2	3	1
b)	Explain the structure and components of an X.509 certificate format with a neat diagram.	8	L2	4	1
6. a)	Explain the Elgamal cryptosystem.	6	L2	3	1
b)	Explain the Generic Identity Management Architecture.	6	L2	4	1
c)	Explain the process of distribution of public keys using the Public-key certificate scheme.	4	L2	4	1

Unit – III

7. a)	Explain the ESP packet format with a neat diagram.	8	L2	5	1
b)	Explain the components of the SSL protocol stack.	8	L2	5	1
8. a)	Explain the Anti-replay mechanism in detail with a neat diagram.	8	L2	5	1
b)	Explain the various web security threats along with its consequences and the countermeasures. Also state the relative location of the security facilities in the TCP/IP protocol stack.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
